

Teichmüller theory Problem set 1

1. Let $f : U \subset \mathbb{C} \rightarrow V \subset \mathbb{C}$ be a QC map between domains U, V in $\mathbb{C}$ and let $\mu = \mu_f$. Suppose $\gamma \in \text{Aut}(U)$ is such that $\mu(z) = \mu(\gamma(z)) \cdot \overline{\gamma'(z)}/\gamma'(z)$ for a.e. $z \in U$. Show that $f \circ \gamma \circ f^{-1} \in \text{Aut}(V)$.
2. Let $X_i = \mathbb{H}/\Gamma_i, i = 1, 2$ be Riemann surfaces, let $f : X_1 \rightarrow X_2$ be a QC homeomorphism, let $F : \mathbb{H} \rightarrow \mathbb{H}$ be a lift of f , and let $\mu = \mu_F$. Show that for all $\gamma \in \Gamma_1$, $\mu(z) = \mu(\gamma(z)) \cdot \overline{\gamma'(z)}/\gamma'(z)$ for a.e. $z \in \mathbb{H}$.
3. Let $f_n : \mathbb{C} \rightarrow \mathbb{C}$ be a sequence of K -QC maps. Suppose that $|f_n(0)|, |f_n(1)| < M$ for all n for some $M > 0$, and that $|f_n(0) - f_n(1)| > \epsilon$ for all n , for some $\epsilon > 0$. Show that $\{f_n\}_{n \geq 1}$ has a subsequence which converges uniformly on compacts in $\mathbb{C}$ to a K -QC map $f : \mathbb{C} \rightarrow \mathbb{C}$.
4. Show that any K -QC map $f : \mathbb{D} \rightarrow \mathbb{D}$ extends to a K -QC map $F : \hat{\mathbb{C}} \rightarrow \hat{\mathbb{C}}$ such that $F = r \circ f \circ r$, where $r(z) = 1/\bar{z}$. In particular f extends continuously to a homeomorphism $F : \overline{\mathbb{D}} \rightarrow \overline{\mathbb{D}}$.
5. Let $f_n : \mathbb{D} \rightarrow \mathbb{D}$ be a sequence of K -QC maps such that $|f_n(0)| < 1 - \epsilon$ for all n , for some $0 < \epsilon < 1$. Suppose $f_n|_{S^1} \equiv id$ for all n . Show that $\{f_n\}_{n \geq 1}$ has a subsequence which converges uniformly on compacts in $\mathbb{D}$ to a K -QC map $f : \mathbb{D} \rightarrow \mathbb{D}$.
6. Let $f : \mathbb{D} \rightarrow \mathbb{D}$ be an o.p. homeomorphism which is L -Lipschitz with respect to the hyperbolic metric on $\mathbb{D}$. Show that f is K -QC where $K = L^2$.
7. For a Riemann surface X with universal cover $\mathbb{D}$, let g_X denote the unique complete conformal metric on X of constant curvature -1 .
Let X, Y be Riemann surfaces X, Y with universal cover $\mathbb{D}$.
 - a) If there is an o.p. diffeomorphism $f : X \rightarrow Y$, show that f is biholomorphic iff $f : (X, g_X) \rightarrow (Y, g_Y)$ is an isometry.
 - b) If there is a holomorphic covering map $\pi : X \rightarrow Y$, then show that $\pi^*g_Y = g_X$ and so $\pi : (X, g_X) \rightarrow (Y, g_Y)$ is a local isometry.
8. Let $\gamma_n \in \text{Aut}(\mathbb{D}), n \geq 1$. Suppose $\gamma_n(z_0) \rightarrow \xi \in S^1$ as $n \rightarrow \infty$ for some $z_0 \in \mathbb{D}$ and some $\xi \in S^1$. Show that $\gamma_n(z) \rightarrow \xi$ as $n \rightarrow \infty$ uniformly on compacts in $\mathbb{D}$.
9. Let X, Y be compact Riemann surfaces of genus $g \geq 2$. Show that two QC maps $f, g : X \rightarrow Y$ are homotopic iff there exist lifts $\tilde{f}, \tilde{g} : \mathbb{D} \rightarrow \mathbb{D}$ such that $\tilde{f} = \tilde{g}$ on S^1 .
10. Let L be a lattice in $\mathbb{C}$ and let $X = \mathbb{C}/L$. For $c \in \mathbb{C}$ let $T_c \in \text{Aut}(X)$ be the map $T_c : X \rightarrow X, z + L \mapsto z + c + L$.
 - a) For $f \in \text{Aut}(X)$, show that f is homotopic to the identity iff $f = T_c$ for some $c \in \mathbb{C}$.
 - b) For $f \in \text{Aut}(X)$, show that f^n is homotopic to the identity for some $n \geq 1$.
11. Let X be a compact Riemann surface of genus $g \geq 2$. Show that $f \in \text{Aut}(X)$ is homotopic to the identity iff $f = id$.